KCC CODE CLUB · GAME MAKERS

Week 5

Variables and Score

Week 5 of 8

Name _____

Today's big idea

A **variable** is a box with a label on it. You put a number inside. You can look at what's in it, and you can change it while the game is running.

```
┌─────────┐
| SCORE |
| 7 |
└─────────┘
```

`set` replaces what's in the box. `change` adds to what's already there.

`set score to 5` → the box now holds 5, whatever it held before. `change score by 1` → if it held 5, it now holds 6.

Words to know

Word	What it means
variable	A named box that holds a value
set	Replace what's in the box
change	Add to what's in the box (a negative number subtracts)
reset	Set your variables back to their starting values when the game begins

> **Always reset.** Put `set score to 0` under the green flag, or your next game will start with your last score still in the box.

Class activity — Countdown Timer

```
when green flag clicked                    EVENTS
set timeleft to 30                         VARIABLES
```

```
repeat 30                                   CONTROL
    wait 1 seconds
    change timeleft by -1                   VARIABLES
say [Time's up!]                            LOOKS
```

Watch it count down. Then try changing the 30 — what else has to change with it?

Your project — keeping score

- [ ] Make a variable called `score`
- [ ] `set score to 0` under the green flag
- [ ] Good catch → `change score by 1`
- [ ] Bad catch → `change score by -1` and a different sound
- [ ] Put the score somewhere you can read it
- [ ] Playtest — then swap and try to beat your neighbour's high score
- [ ] **Save it**

Homework

1. High score. Add a `highscore` variable that only updates when `score` beats it.

```
if <(score) > (highscore)> then             CONTROL
    set highscore to (score)                VARIABLES
```

Careful: do NOT reset `highscore` under the green flag, or it isn't a high score!

2. Make it look good. Position your score display sensibly. Try right-clicking it for a large readout. Hide any variables the player doesn't need to see.

Bonus challenges

1. Make the game get faster as your score goes up — put `score` into the `wait` in the falling loop. *(Careful: a wait can't be negative.)*
2. Add a `lives` variable starting at 3 that goes down on a bad catch. What should happen at zero? See if you can make something happen. *(That's next week — nice head start!)*
3. **Click Counter.** `set clicks to 0` on the green flag, `change clicks by 1` when the sprite is clicked, and `say [You win!]` at 10. Then add your countdown timer — how many clicks can you get in 30 seconds? That's a whole little game!

4. Add a `level` variable that goes up every ten points.

If something breaks

- **Score jumps by 10 or 20 per catch?** You're still touching the object for several loops. Make sure the object jumps back to the top *immediately* inside the same `if`.
- **Score starts at last game's number?** Missing `set score to 0` under the green flag.
- **High score keeps resetting?** You put `set highscore to 0` under the green flag. Take it out.

Stuck? Put your hand up — that is what we are here for. And always save before you close the lid.